



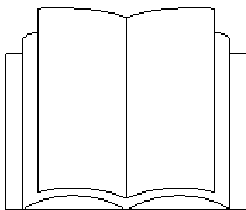
Annex for the original Instruction Manual



Log Cutter/Splitter

JAPA 305

Product number:	300-700-series loaders:	A415435
	200-series loaders:	A415434



Please read carefully the original attachment's instruction manual before using the equipment, as well as this annex, and follow all instructions.

This annex for the operator's manual includes specific information about the use of the equipment with Avant multi purpose loaders.

Keep the original operator's manual as well as this annex for later reference.

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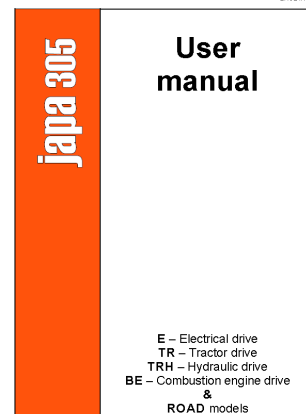
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Annex: JAPA 305 User Manual



This annex for the original User Manual of the log cutter-splitter does include all necessary information about safe and correct use of the equipment. Rnsure that you have received and read also the original User Manual, which is normally shipped with the machine. Contact your sales representative or the manufacturer immediately, if the original User Manual is not available.



1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for AVANT multi purpose loaders. The log cutter/splitter has been adapted for use with AVANT loaders by the original manufacturer of the equipment. The original User Manual, provided with the log splitter/cutter, includes important information about safe use of the equipment. This annex for the manual includes important information of coupling and using the equipment with AVANT loaders.

By familiarising yourself with and following the instructions shown in the original User Manual and the additional information in this annex, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Ensure that all persons using the loader and the log cutter/splitter have received proper guidance and familiarised themselves with the manuals of the loader and attachment and all safety instructions before using the equipment.

Keep this annex of the manual and the original user manual at hand throughout the service life of the equipment. If this annex for the manual or the original operator's manual is lost or damaged, you can request a new one from your nearest AVANT dealer. If you sell or transfer the equipment, be sure to hand both parts of the manual to the new owner.

This manual uses the following warning symbols to indicate important points that must be taken into account to reduce the risk of personal injury or property damage:

	WARNING: This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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Note:

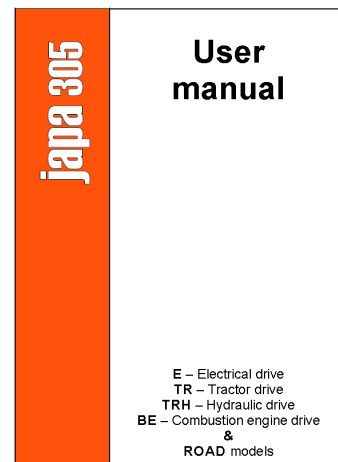
This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This annex for the instruction manual is original English version.



This annex for the original instruction manual of the log cutter/splitter machine does not cover all necessary information needed for safe operation of the equipment. This annex is not intended to be distributed separately. The original User Manual, provided with the machine, should be considered as the main source of information about safety, use, technical information, and the maintenance of the equipment. This annex includes important information about the use of the equipment specifically with AVANT multi purpose loaders.



2. Designed purpose of use

The JAPA 305 Avant model firewood machines are hydraulically operated attachments designed and manufactured to be suitable for use with 200-700 –series AVANT multi purpose loaders. They are intended to be used by one person at a time for cutting firewood and splitting it into halves or quarters. The log cutter/splitter cuts the log with a chain saw blade to the adjusted length, and then the log is split with a hydraulic cylinder, which presses the log against a splitting wedge. Split logs will get pushed automatically out, or the optional conveyor system can be used.

The log cutter/splitter is intended to be used only when it is securely locked to the quick coupling plate of AVANT loader. The attachment is powered by the hydraulic outlet of the Avant loader. The log conveyor, shown in the original manual, is optional extra. Contact your AVANT dealer for further information about its availability.

Familiarise yourself with the instruction manual's instructions regarding service and maintenance. Demanding repair and maintenance operations, such as hydraulic repairs, should be left for professional maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for using the log splitter with Avant loaders

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



Misuse or careless use, or using equipment that is in bad condition, may cause serious injuries. Get to know the controls of the loader and coupling procedures of the attachment at a safe area. In particular, familiarize yourself with procedures how to stop the equipment in a safe manner and to make it safe. Read the following safety precautions carefully.



- When attaching the attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are in the lower position and that they have locked the attachment to the loader.**
- The log cutter/splitter is to be used by one operator at a time. Keep others at a safe distance, and do not let others to operate the controls of the loader while using the attachment.
- In case of fault or blockage, make sure that the equipment has been made safe by turning off the engine of the loader. When a blockage is released the moving parts of the attachment may move unexpectedly, causing the saw chain to move or pushing of the splitting cylinder. Follow the safe stopping procedure before handling the chainsaw or when reaching to the crushing hazard area of the splitting cylinder.
- When transporting the equipment, pay attention to the contours of the terrain as well as other hazards, such as loose rocks or branches and trees reaching the driver's area.
- Always transport the log cutter/splitter as low and close to the machine as possible to keep the centre of gravity low. Keep the telescopic boom retracted and drive calmly.
- Keep hands and feet away from moving parts.
- When uncoupling the log cutter/splitter from the loader, ensure that it is secured against accidental movement.
- Never perform any maintenance or adjustments while the loader engine is running. Also, release the residual hydraulic pressure of the system.
- Do not modify the structure of the attachment in a manner that would affect its safety. Check that all safety features shown in the original User manual are in working condition.
- Also read the safety instructions for the loader in the loader manual. Get to know the loader controls and how to operate it safely.

Safe stopping of the equipment, before going near the blade or splitting area:



- Stop the attachment by turning its control lever to position "off" and turn off auxiliary hydraulics of the loader.
- Make sure that the log machine is firmly against the ground and will remain stable.
- Shut down the loader engine and lock the parking brake.
- Release residual pressure from the hydraulic system by moving the auxiliary hydraulics control lever to its extreme positions.
- Remove key from ignition to make sure that another person can't operate the controls of the loader.



Note that the loader boom moves down and the log machine can be tilted even if the engine has been stopped. Never go under or in front of raised boom or attachment. Always lower the attachment on the ground before leaving the driver's seat.



Note the effect of heavy load to the the stability and handling characteristics and of the loader, especially when operating with the smaller loader models. Always keep the load close to the ground and keep the telescopic boom retracted. Drive slowly and carefully and avoid steep slopes. Use additional counterweights as needed. Read the operator's manual of the loader.



Remember to wear proper protective clothing:



- The noise level at the operator's area may exceed 85 dB(A). Wear hearing protection.
- Wear safety boots while working with the loader.



- Wear protective gloves. and when handling hydraulic. Wear also safety goggles when handling hydraulic hoses or components.

4. Technical specifications

The labels and markings, which must be visible on the equipment, are listed in the original User Manual. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer.

Table 1 –Log cutter/splitter specifications

Product number	A415434	A415435
Log length, min-max:	200 - 500 mm	
Log diameter, min-max:	20 - 300 mm	
Weight:	265 kg	
Minimum operating temperature:	-25°C	
Saw flange:	13"	
Splitting force:	30 kN	
Dimensions, transport configuration: L x W x H:	2780 x 1030 x 2450 mm	
Maximum input of hydraulic energy:	30 l/min, 22,5 MPa (225 bar)	
Suitable AVANT loader:	216, 218, 220	320-750

5. Assembly of the log cutter/splitter

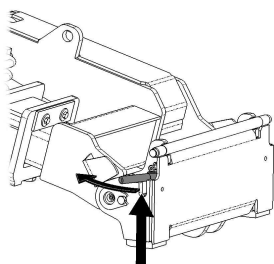
The log cutter/splitter is ready for use as delivered. Attach the attachment and its hydraulic hoses according to the following instructions.

5.1. Coupling the log cutter/splitter to the loader

The attachment is mounted with the quick attach plate of the loader boom and the counterpart on the attachment, which are not shown in the original User Manual. The locking is semi-automatic, but due to the relatively high position of the quick attach plate on the log splitter, and the risk of tipping over of the attachment, the locking pins must always be locked manually.

Coupling the attachment to the loader is quick and easy, but it must be done carefully. The attachment is fixed to the loader boom by using the quick attach plate and the counterpart on the attachment. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must not be raised over one metre under any circumstance if it has not been locked. To prevent hazardous situations, always follow the attachment coupling instructions provided in the following pages. Also remember the safety instructions shown in Chapter 3 and additional information shown in the operator's manual of the loader. The attachment is mounted to the loader as follows:

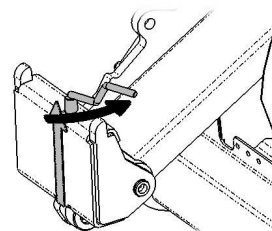
300-700 series loaders:



Step 1:

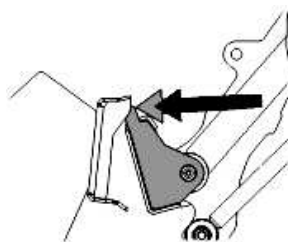
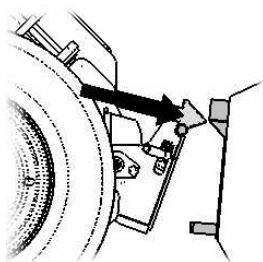
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.

200 series loaders:



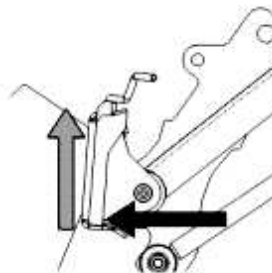
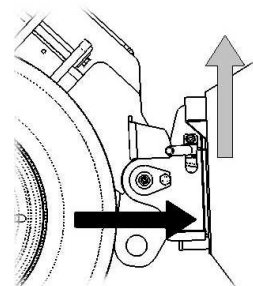
Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this. Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.



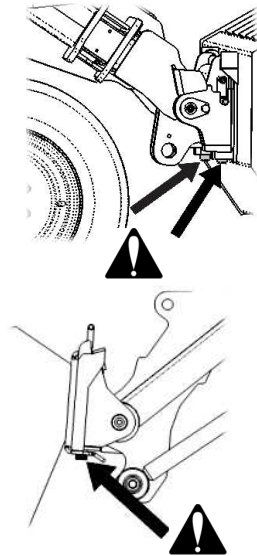
Step 3:

- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment. Lock the
- Lock the locking pins manually or lock the hydraulic locking. Always check the locking of pin / pins.





An attachment that has not been completely locked to the loader may fall on the boom or operator, or under the loader during driving, and cause serious injuries or loss of control of the loader. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or lift an attachment that has not been locked.



Do not use the automatic locking of the locking pins with the log cutter/splitter, because the quick attach plate of the machine is at high level, and tilting of an unlocked log cutter/splitter increases the risk of toppling over. It is recommended to lower the attachment on the ground and secure the locking manually, as it is necessary to leave the driver's seat to connect the hydraulic couplings.

5.2. Connecting and disconnecting the hydraulic fittings

On 300-700 series loaders hydraulic hoses are connected using the multiconnector system. . If you want to change the type of the hydraulic couplings, contact your Avant dealer or service point for instructions or installation services. On 200-series loaders only conventional quick couplings are available.



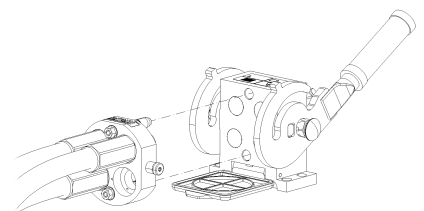
Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries or burns.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



Connecting conventional quick couplings:

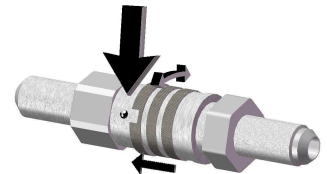
Before connecting or disconnecting the standard quick couplings, the residual pressure must be released as shown below. The conventional quick couplings will not connect, if there is pressure in the hydraulic system.

To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Make sure that both fittings are properly connected, otherwise the hydraulic motor of the mower may get damaged.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. There is a small groove in the collar of all female fittings, and a small ball near the collar. When connecting or disconnecting, the groove and ball must be aligned. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The collar may also turn accidentally to the locking position, making it impossible to connect or disconnect the couplings.



Releasing residual hydraulic pressure:

To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader to its extreme positions before disconnecting the couplings. If the fittings will not connect to the loader, the residual pressure must be released. The residual pressure must be released before handling the blades of the attachment and before storage of the attachment.

Disconnecting the hydraulics:

Before disconnecting the fittings, place the attachment on solid and level surface, as described in following paragraph. Turn the control lever of the auxiliary hydraulics to its neutral position.



When uncoupling the attachment from the loader, always disconnect the hydraulic fittings first before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent dirt from entering the hydraulic system.

6. Instructions for use

Once the log cutter/splitter and its hydraulic hoses have been coupled according to the instructions, it is ready for work. See the original operator's manual of the equipment for more information about operating the machine as well as the following additional information.



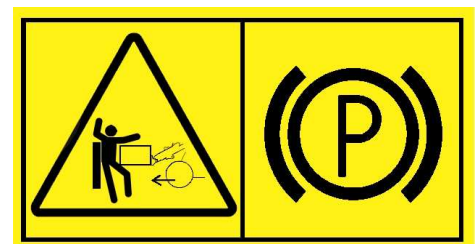
In addition to this annex of the manual, also read the original User manual for correct adjustments, operation and maintenance of the equipment, or cleaning of the equipment if the splitter can start by accident. Shut down the loader engine first.



Never perform any adjustments, maintenance or cleaning of the equipment if the splitter can start by accident. Follow safe stopping procedure to ensure that the machine can't start by accident.

6.1. Setting up the log cutter/splitter

- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. The equipment must not be used, if there is a fault in the hydraulic system of the loader or the attachment. Keeping the splitter clean will reduce the possibility of blockages.
- Transport the log cutter/splitter carefully and note the effect of weight to the stability of the loader.
- Choose a level working site, which can carry the load of the equipment and the logs.
- The operating area must be even and it must not be slippery. Clear the area from any obstacles that could cause a tripping hazard.
- Operate the log cutter/splitter in daylight or outdoor area with good artificial light.
- Check regularly to ensure there are no hydraulic leaks. Refer to original instruction manual for maintenance instructions.
- Ensure that the loader and the attachment are used in a safe and intended manner. Smoking near the operating area is prohibited. Do not allow children to operate the equipment.
- Make sure to lower the attachment securely on the ground. Ensure that the parking brake of the loader is engaged and that the splitter can not move during use.



6.2. Use



Start the machine only after it has been set to working position and work can be started. Always turn off the auxiliary hydraulics when the attachment is not in use. Do not leave the machine unattended when the engine of the loader is running. Follow safe stopping procedure always, when handling the chainsaw or when reaching the splitting area is necessary for cleaning, splitter wedge height adjustment or when removing blockages. The attachment is intended to be used by only one person at a time.



When operating the log splitter the loader must be equipped with a hydraulic oil cooler. Without the cooler hydraulic oil may overheat. Overheating of the oil decreases power and affects the service life of the hydraulic components of both the attachment and the loader. Clean the loader oil cooler core regularly; refer to the loader manual for more instructions.

To start the log cutter/splitter, turn the auxiliary hydraulics lever on the loader to its locked position and set moderate engine rpm, depending on the working pace. Depending on the loader model, two hydraulic pumps may be used to reduce engine noise. Do not exceed the maximum permissible input of hydraulic oil; full rpm must not be used when operating with 600 or 700 series loaders. Note that unnecessarily high engine rotating speed will not improve the splitting force. Operate the log cutter/splitter and perform maintenance procedures as instructed in the original manual of the equipment, so that the performance of the machine will remain at high level.



Check the operation and the adjustments of the splitter with a few empty cycles before loading a log.

The lubrication of the log cutter chain saw consumes a small amount of hydraulic oil from the loader. Check the hydraulic oil level regularly.



Do not use the attachment for any other purpose than for cutting and splitting firewood.

7. Maintenance and warranty

Instructions for maintenance and service are provided with the original instruction manual as well as the spare parts list. Contact your AVANT dealer for parts and service.

The manufacturer of the log cutter/splitter grants the warranty of one year for the log cutter/splitter. Warranty terms are provided with the instruction manual. For warranty claims, contact your AVANT dealer for further information.

The manufacturer of the log cutter/splitter provides the declaration of conformity for the equipment.